



FINSHIELD

UPI Fraud Transaction Analytics

1. Executive Summary

FinShield is an end-to-end UPI fraud analytics project designed to turn transaction-level data into management-ready fraud monitoring decisions. The project examines four connected areas: fraud scale and financial exposure, transaction/device/channel behavior, risk and location signals, and time/operational monitoring.

The analysis starts with a transaction-level dataset. The data was inspected and cleaned before analysis. Python was then used for exploratory data analysis, SQL for structured and repeatable business questions, and Power BI for an interactive three-page fraud monitoring dashboard.

Core Business Problem

How can a digital payments business identify transaction, device, location, behavioral, and risk characteristics associated with fraudulent UPI activity and use them to strengthen fraud monitoring?

Headline Business KPIs

KPI	Result Shown in Dashboard	Why Management Cares
Total Transactions	50,000	Measures transaction volume.
Fraudulent Transactions	2,794	Shows the observed fraud volume.
Fraud Rate	5.59%	Measures the share of transactions labeled fraudulent.
Total Transaction Amount	≈ ₹40.77 Million	Shows the financial value processed in the dataset.
Fraudulent Transaction Amount	≈ ₹2.23 Million	Shows transaction value associated with fraud labels.
Average Transaction Amount	≈ ₹815.48	Provides the overall transaction-size baseline.
Average Fraud Amount	≈ ₹798.84	Compares transaction size within fraud-labeled activity.

Fraud-labeled transactions represent 5.47% of total transaction value. The fraud-labeled average transaction amount (₹798.84) is close to the overall average (₹815.48), so transaction amount alone does not clearly separate fraud from legitimate activity.

2. Raw Data Understanding

The raw dataset contains 50,000 transaction-level records and 21 fields. Each row represents a UPI transaction and combines transaction, participant, bank, location, device, behavioral, risk, and fraud-label information.

Field Group	Important Columns	Business Role
Transaction	Transaction_ID, Transaction_Date, Transaction_Type, Amount_INR, Transaction_Status	Describes the transaction, timing, type, amount and processing status.
Participants	Sender_ID, Receiver_ID, Sender_Bank, Receiver_Bank	Identifies sending/receiving entities and banking relationships.
Location / Device	City, Device_Type, Location_Change_KM	Supports geographic and device-behavior analysis.
Behavior	Transactions_Last_24H, Avg_Transaction_Amount_7D, Failed_Transactions_24H, Account_Age_Days	Captures recent activity and account context.
Risk	IP_Risk_Score, Device_Risk_Score	Provides risk indicators for transaction monitoring.
Fraud Labels	Fraud_Reason, Fraud_Label, Is_Fraud	Defines the supplied fraud classification and reason.

Raw Data Quality Questions

- Are transaction IDs unique and usable?
- Are dates correctly typed and numeric measures stored as numeric values?
- Are missing or duplicate transaction records present?
- Are categorical values consistent across banks, cities, devices and transaction types?
- Are amount, risk, activity and location measures within reasonable business ranges?
- Could unusual values be genuine transaction behavior rather than data-entry errors?

3. Data Cleaning & Preparation

Cleaning was performed with the objective of making the fraud dataset analytically reliable without destroying legitimate unusual behavior. For each issue, the project considered the affected field, the business consequence, and the appropriate treatment.

Area / Column	Issue Checked	Treatment / Decision	Why It Matters
All rows	Missing values / duplicate rows	Validated the supplied cleaned file; no missing cells or duplicate full rows remained.	Prevents inflated or incomplete fraud KPIs.
Transaction_ID	Duplicate / unusable IDs	Checked uniqueness; transaction IDs were unique.	Supports reliable transaction-level analysis.
Transaction_Date	Wrong type / invalid dates	Converted and validated as datetime.	Required for hourly/monthly analysis.
Categorical fields	Inconsistent text	Checked category consistency before grouping.	Prevents fragmented segment results.
Amount_INR	Extreme transaction values	Investigated statistical outliers using business logic.	A genuine large transaction can be a useful fraud signal.
IP_Risk_Score	Values outside risk range	Checked against 0–100 business range.	Required for risk segmentation.
Device_Risk_Score	Values outside risk range	Checked against 0–100 business range.	Required for risk segmentation.
Location_Change_KM	Large statistical values	Investigated rather than automatically deleting.	Large movement can be a genuine behavioral signal.
Transactions_Last_24H / Failed_Transactions_24H	High activity values	Investigated as behavioral signals.	Unusual activity may be relevant to fraud detection.
Fraud_Reason	Label/reason mismatch	1,987 fraud-labeled rows have 'Normal Activity'; retained but flagged.	Should be resolved before using reason as a fully reliable driver.

Important Cleaning Principle

The project did not follow the rule “remove every outlier.” An unusual value was first evaluated against business logic. For example, a high transaction amount or large location change may be a genuine event, while an invalid range or inconsistent fraud reason requires data-quality attention. This distinction is important because removing genuine unusual behavior can hide the exact signals a fraud-monitoring team needs to investigate.

4. Python Exploratory Data Analysis (EDA)

Python was used after cleaning to explore the dataset and identify the KPIs and patterns that deserved deeper analysis in SQL and Power BI. The focus was on a small set of business-relevant analyses rather than producing unrelated visualizations.

4.1 Fraud Scale

```
total_transactions = len(df)
fraud_transactions = df['Is_Fraud'].sum()
fraud_rate = df['Is_Fraud'].mean() * 100
```

Why used: Establishes the fraud baseline. The dataset contains 50,000 transactions, 2,794 fraud-labeled transactions and a 5.59% observed fraud rate.

4.2 Fraudulent Transaction Amount

```
fraud_amount = df.loc[df['Is_Fraud'] == 1, 'Amount_INR'].sum()
```

Why used: Measures the transaction value associated with fraud labels and adds a financial-exposure view to the simple fraud-count KPI.

4.3 Average Fraud vs Legitimate Amount

```
avg_fraud = df.loc[df['Is_Fraud'] == 1, 'Amount_INR'].mean()
avg_legit = df.loc[df['Is_Fraud'] == 0, 'Amount_INR'].mean()
```

Why used: Tests whether transaction amount alone separates fraud from legitimate activity. In this dataset, the averages are close.

4.4 Fraud by Transaction Type

```
type_fraud_rate = df.groupby('Transaction_Type')['Is_Fraud'].mean() * 100
```

Transaction Type	Observed Fraud Rate
Collect Request	7.40%
UPI ID	5.73%
QR Code	5.64%
Mobile Number	5.58%

Why used: Identifies transaction channels where fraud labels are more concentrated. Collect Request has the highest observed rate in the dataset.

4.5 Fraud by Device Type

```
device_fraud_rate = df.groupby('Device_Type')['Is_Fraud'].mean() * 100
```

Device Type	Observed Fraud Rate
Web	9.13%
Android	5.41%
iOS	5.38%

Why used: Supports device-focused monitoring and helps compare observed fraud concentration across access methods.

4.6 Fraud by City

```
city_fraud_rate = df.groupby('City')['Is_Fraud'].mean() * 100
```

City	Observed Fraud Rate
Bhopal	6.09%
Kolkata	5.99%
Hyderabad	5.86%
Bangalore	5.76%
Chennai	5.67%
Pune	5.62%
Delhi	5.59%
Jaipur	5.55%
Ahmedabad	5.54%
Mumbai	5.37%

Why used: Provides geographic context. These are observed associations in the project dataset and should not be interpreted as evidence that a city causes fraud.

4.7 Risk Score Analysis

Risk scores were grouped into monitoring bands to make the relationship between score level and observed fraud rate easier to interpret.

Risk Band	IP Risk Fraud Rate	Device Risk Fraud Rate
0-20	2.52%	2.51%
21-40	3.37%	3.42%
41-60	4.68%	4.62%
61-80	6.56%	6.42%
81-100	10.44%	9.53%

5. Python EDA — Risk, Location & Time

5.1 Location Change

```
location_band = pd.cut(df['Location_Change_KM'], [-1,5,10,20,40,100],
                        labels=['0-5','6-10','11-20','21-40','41+'])
location_rate = df.groupby(location_band, observed=True)['Is_Fraud'].mean() * 100
```

Location Change Band	Observed Fraud Rate
0-5 km	5.52%
6-10 km	5.50%
11-20 km	5.68%
21-40 km	5.99%
41+ km	9.07%

Why used: Large location changes can represent unusual movement. The 41+ km band shows a higher observed fraud rate and can be combined with other risk signals for investigation.

5.2 Fraud by Transaction Hour

```
df['Hour'] = df['Transaction Date'].dt.hour
hour_fraud_rate = df.groupby('Hour')['Is_Fraud'].mean() * 100
```

Why used: Time-based analysis helps identify periods where monitoring intensity may need to change. In the dataset, the highest observed hourly fraud rate is 6.92% at 20:00, followed by 6.71% at 02:00.

5.3 Fraud by Month

```
df['Month'] = df['Transaction Date'].dt.month
monthly_fraud_rate = df.groupby('Month')['Is_Fraud'].mean() * 100
```

Why used: Monthly analysis provides a recurring trend view and can be used to monitor whether the fraud baseline is changing over time.

5.4 High-Risk Segment

```
high_risk = (df['IP Risk Score'] >= 81) | (df['Device Risk Score'] >= 81)
high_risk_rate = df.loc[high_risk, 'Is_Fraud'].mean() * 100
```

Why used: A simple high-risk segment provides an operational starting point for enhanced review. It should be treated as a monitoring rule, not as proof that every high-risk transaction is fraudulent.

Python EDA → Business Value

Finding Area	Business Question	Decision Supported
Fraud Scale	How much fraud is present?	Set the monitoring baseline.
Fraud Amount	What transaction value is associated with fraud?	Understand financial exposure.
Transaction Type	Which transaction types show higher observed fraud?	Focus channel monitoring.
Device	Which device types show higher observed fraud?	Review access-channel risk.
Risk Scores	Does fraud rate change across risk bands?	Support risk-based monitoring.
Location	Does large movement coincide with higher fraud?	Investigate unusual movement.
Time	When does observed fraud concentration change?	Support time-based monitoring.

6. SQL Business Analysis

SQL was used to convert exploratory findings into structured, reproducible business analysis. The key advantage is that SQL can calculate KPIs and segment the transaction population consistently for reporting and dashboard use.

6.1 Total Transactions & Fraud Rate

```
SELECT
    COUNT(*) AS Total_Transactions,
    SUM(Is_Fraud) AS Fraudulent_Transactions,
    100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions;
```

Why this KPI is useful: Creates the primary fraud baseline against which all segments can be compared.

6.2 Total Transaction Amount & Fraud Amount

```
SELECT
    SUM(Amount_INR) AS Total_Transaction_Amount,
    SUM(CASE WHEN Is_Fraud = 1 THEN Amount_INR ELSE 0 END)
    AS Fraudulent_Transaction_Amount
FROM UPI_Transactions;
```

Why this KPI is useful: Adds financial exposure to the transaction-count view.

6.3 Fraud Rate by Transaction Type

```
SELECT
    Transaction_Type,
    COUNT(*) AS Total_Transactions,
    SUM(Is_Fraud) AS Fraud_Count,
    100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY Transaction_Type
ORDER BY Fraud_Rate_Pct DESC;
```

Why this KPI is useful: Identifies transaction types requiring closer monitoring.

6.4 Fraud Rate by Device Type

```
SELECT
    Device_Type,
    COUNT(*) AS Total_Transactions,
    SUM(Is_Fraud) AS Fraud_Count,
    100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY Device_Type
ORDER BY Fraud_Rate_Pct DESC;
```

Why this KPI is useful: Compares fraud concentration across access devices.

6.5 Fraud Rate by City

```
SELECT
    City,
    COUNT(*) AS Total_Transactions,
    SUM(Is_Fraud) AS Fraud_Count,
    100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY City
ORDER BY Fraud_Rate_Pct DESC;
```

Why this KPI is useful: Provides geographic monitoring context.

6.6 Fraud Rate by IP Risk Range

```
SELECT
    CASE
        WHEN IP_Risk_Score <= 20 THEN '0-20'
        WHEN IP_Risk_Score <= 40 THEN '21-40'
        WHEN IP_Risk_Score <= 60 THEN '41-60'
        WHEN IP_Risk_Score <= 80 THEN '61-80'
        ELSE '81-100'
    END AS IP_Risk_Range,
    COUNT(*) AS Transactions,
    SUM(Is_Fraud) AS Fraud_Count,
    100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY
    CASE
        WHEN IP_Risk_Score <= 20 THEN '0-20'
        WHEN IP_Risk_Score <= 40 THEN '21-40'
        WHEN IP_Risk_Score <= 60 THEN '41-60'
        WHEN IP_Risk_Score <= 80 THEN '61-80'
        ELSE '81-100'
    END;
```

Why this KPI is useful: Converts a continuous risk score into a management-friendly monitoring range.

6.7 Fraud Rate by Device Risk Range

```
SELECT
    CASE
        WHEN Device_Risk_Score <= 20 THEN '0-20'
        WHEN Device_Risk_Score <= 40 THEN '21-40'
        WHEN Device_Risk_Score <= 60 THEN '41-60'
        WHEN Device_Risk_Score <= 80 THEN '61-80'
        ELSE '81-100'
    END AS Device_Risk_Range,
```

```

COUNT(*) AS Transactions,
SUM(Is_Fraud) AS Fraud_Count,
100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY
CASE
    WHEN Device_Risk_Score <= 20 THEN '0-20'
    WHEN Device_Risk_Score <= 40 THEN '21-40'
    WHEN Device_Risk_Score <= 60 THEN '41-60'
    WHEN Device_Risk_Score <= 80 THEN '61-80'
    ELSE '81-100'
END;

```

Why this KPI is useful: Tests whether higher device-risk scores are associated with higher fraud rates.

6.8 Location Change Analysis

```

SELECT
CASE
    WHEN Location_Change_KM <= 5 THEN '0-5'
    WHEN Location_Change_KM <= 10 THEN '6-10'
    WHEN Location_Change_KM <= 20 THEN '11-20'
    WHEN Location_Change_KM <= 40 THEN '21-40'
    ELSE '41+'
END AS Location_Change_Range,
COUNT(*) AS Transactions,
SUM(Is_Fraud) AS Fraud_Count,
100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY
CASE
    WHEN Location_Change_KM <= 5 THEN '0-5'
    WHEN Location_Change_KM <= 10 THEN '6-10'
    WHEN Location_Change_KM <= 20 THEN '11-20'
    WHEN Location_Change_KM <= 40 THEN '21-40'
    ELSE '41+'
END;

```

Why this KPI is useful: Supports investigation of unusually large location changes.

6.9 Fraud by Transaction Hour

```

SELECT
DATEPART(HOUR, Transaction_Date) AS Transaction_Hour,
COUNT(*) AS Transactions,
SUM(Is_Fraud) AS Fraud_Count,
100.0 * SUM(Is_Fraud) / COUNT(*) AS Fraud_Rate_Pct
FROM UPI_Transactions
GROUP BY DATEPART(HOUR, Transaction_Date)
ORDER BY Transaction_Hour;

```

Why this KPI is useful: Supports time-based monitoring and operational scheduling.

7. Power BI Dashboard & DAX

Power BI was used as the final decision-support layer. The dashboard does not simply repeat every analysis. Instead, it groups the fraud analysis into three management-focused pages and uses reusable measures so KPIs can respond to filters and slicers.

7.1 DAX Measures

Measure	DAX Formula	Why Used
Total Transactions	Total Transactions = COUNTROWS(UPI_Transactions)	Core transaction KPI.
Total Fraud Transaction	Total Fraud Transaction = CALCULATE(COUNTROWS(UPI_Transactions), UPI_Transactions[Is_Fraud] = 1)	Counts fraud-labeled transactions.
Fraud Rate %	Fraud Rate % = DIVIDE([Total Fraud Transaction], [Total Transactions], 0)	Measures fraud incidence.
Total Fraud Amount	Total Fraud Amount = CALCULATE(SUM(UPI_Transactions[Amount_INR]), UPI_Transactions[Is_Fraud] = 1)	Measures financial exposure.
Average Fraud Amount	Average Fraud Amount = CALCULATE(AVERAGE(UPI_Transactions[Amount_INR]), UPI_Transactions[Is_Fraud] = 1)	Compares fraud transaction size.
Average Transaction Amount	Average Transaction Amount = AVERAGE(UPI_Transactions[Amount_INR])	Provides overall amount baseline.

7.2 Dashboard Page Design

Page	Business Question	Main KPIs / Visuals	Management Value
1. Home	What is the overall fraud picture?	Total Transactions, Total Fraud Transaction, Fraud Rate %, Total Fraud Amount, Average Fraud Amount; fraud vs non-fraud; hourly/monthly trend; device, transaction type and city.	Provides the management-level fraud overview.
2. Risk Analysis	Which risk signals are associated with fraud?	IP Risk Score vs Device Risk Score, IP Risk Range, Device Risk Range, transaction-frequency range, device/type and city/type breakdowns.	Supports risk-based monitoring.
3. Transaction & Location	Where and when is fraud activity concentrated?	Location-change ranges, fraud amount by hour, device/type/city views, risk-vs-amount and location-vs-amount scatter plots.	Supports operational investigation by time, location and behavior.

8. Business Problems Solved

Business Problem	Analysis Used	How the Dashboard Helps
Fraud scale is unclear	Total Transactions, Fraud Count, Fraud Rate	Provides an immediate fraud baseline.
Financial exposure is unclear	Total Transaction Amount, Fraud Amount	Shows transaction value associated with fraud labels.
Risky transaction types are unclear	Fraud Rate by Transaction Type	Highlights transaction types requiring closer review.
Device risk is unclear	Fraud Rate by Device Type and Device Risk Score	Supports device-focused monitoring.
Location risk is unclear	Fraud Rate by City and Location Change	Highlights geographic and movement-related patterns.
Risk-score relationship is unclear	IP Risk Range and Device Risk Range	Shows how observed fraud rate changes across risk bands.
Time concentration is unclear	Fraud Rate by Hour and Month	Supports time-based monitoring.
Behavioral signals are unclear	Transactions Last 24H and Failed Transactions 24H	Allows investigation of unusual recent activity.
Fraud reason quality is unclear	Fraud label vs Fraud reason consistency	Flags a data-quality issue before automated interpretation.

The project therefore addresses the business objective: measure the scale of fraud, understand patterns, identify higher-risk segments, and provide a monitoring-oriented dashboard.

9. Business Insights

- The dataset contains 50,000 transactions, of which 2,794 are labeled fraudulent, producing a 5.59% observed fraud rate.
- Fraud-labeled transactions account for approximately ₹2.23M, or 5.47% of total transaction value.
- Collect Request has the highest observed fraud rate among the listed transaction types at 7.40%.
- Web transactions show a 9.13% observed fraud rate, higher than Android and iOS in this dataset.
- Fraud rate increases across IP risk bands, from 2.52% at 0–20 to 10.44% at 81–100.
- Fraud rate also increases across device-risk bands, from 2.51% at 0–20 to 9.53% at 81–100.
- Transactions with location change above 40 km show a 9.07% observed fraud rate, compared with 5.52% at 0–5 km.
- Bhopal has the highest city-level observed fraud rate among the listed cities at 6.09%.
- The average fraud transaction amount (₹798.84) is close to the legitimate average (₹816.52), so amount alone is not a strong discriminator.
- A data-quality issue remains visible: 1,987 fraud-labeled transactions carry the reason 'Normal Activity'.

10. Business Recommendations

1. Strengthen risk-based monitoring: prioritize transactions with high IP or device risk scores because observed fraud rates are higher in the upper risk bands.
2. Monitor web transactions more closely: the Web device segment has a 9.13% observed fraud rate in the dataset.
3. Review Collect Request activity: its 7.40% observed fraud rate is the highest among the listed transaction types.
4. Investigate large location changes: the 41+ km segment shows a higher observed fraud rate and can be combined with other signals.
5. Use time-based monitoring: hourly analysis shows variation in observed fraud rates, supporting time-aware review rules.
6. Do not rely on transaction amount alone: fraud and legitimate average amounts are close, so amount should be combined with behavioral and risk signals.
7. Resolve fraud-reason data quality: investigate why 1,987 fraud-labeled records are marked 'Normal Activity' before using Fraud_Reason for automated decisions.
8. Use Power BI as a recurring monitoring system: track fraud rate, fraud amount, risk bands, device type, transaction type, city, location change and time patterns.

11. Monitoring Framework

Monitoring Area	Suggested KPI / View	Purpose
Fraud Scale	Fraud Rate %, Fraud Count, Fraud Amount	Track overall fraud exposure.
Risk	IP Risk Range, Device Risk Range	Detect concentration in higher-risk segments.
Behavior	Transactions Last 24H, Failed Transactions 24H	Identify unusual recent activity.
Channel	Fraud Rate by Transaction Type / Device	Monitor channel-specific patterns.
Location	Fraud Rate by City / Location Change	Identify geographic and movement-related concentration.
Time	Fraud Rate by Hour / Month	Identify periods requiring enhanced monitoring.
Data Quality	Fraud Label vs Fraud Reason consistency	Prevent incorrect interpretation of fraud drivers.

12. Conclusion

The FinShield UPI Fraud Transaction Analytics project demonstrates how a Data Analyst can move from raw transaction data to a business-focused fraud monitoring framework. The value of the project is not the dashboard alone; it is the complete analytical process behind it.

Data cleaning created a dependable foundation by checking data types, missing values, duplicates, categorical consistency, valid ranges, statistical outliers, and fraud-label quality. Python EDA then established the fraud KPIs and patterns that deserved deeper investigation. SQL transformed those questions into structured, repeatable analyses. Finally, Power BI converted the results into an interactive monitoring system.

The analysis establishes a 5.59% observed fraud rate across 50,000 transactions, with approximately ₹2.23M in fraud-labeled transaction value. The strongest observed associations are with upper IP/device risk bands, Web transactions, Collect Request transactions, and larger location changes. At the same time, transaction amount alone does not clearly separate fraud from legitimate activity.

The final business outcome is a decision-support framework rather than a collection of charts: measure fraud → identify the driver → investigate the segment → take monitoring action → monitor the result.

13. Project Limitations & Data Quality Notes

- The dataset is a project dataset and should not be interpreted as actual RBI/NPCI production transaction data.
- Fraud rates are descriptive associations within the supplied dataset; they do not establish causation.
- The Fraud_Reason field contains an inconsistency where 1,987 fraud-labeled rows are marked 'Normal Activity'.
- Risk thresholds such as 81+ are analytical monitoring bands used for this project and should be calibrated against real-world model performance before operational deployment.
- The dashboard should be refreshed and monitored over time rather than treated as a one-time fraud conclusion.

14. Interview-Ready Project Summary

I developed an end-to-end UPI Fraud Transaction Analytics project to identify fraudulent transaction patterns and improve fraud-risk monitoring.

I first cleaned and validated 50,000 transaction records using Python/Pandas, checking data types, missing values, duplicates, categorical consistency, valid ranges and statistical outliers. I treated unusual values using business logic instead of automatically deleting them, because unusual transaction behavior can itself be a useful fraud signal.

After cleaning, I used Python for EDA and focused on business KPIs such as total transactions, fraud count, fraud rate, fraudulent transaction amount, transaction type, device type, city, risk scores, location change and time patterns.

I then used SQL to answer structured questions such as fraud rate by transaction type, device and city; risk-band analysis; location-change analysis; time-based fraud patterns; and financial exposure from fraud-labeled transactions.

For the final reporting layer, I built a three-page Power BI dashboard: Home, Risk Analysis, and Transaction & Location. I used KPI cards, slicers, risk ranges, trend visuals and scatter plots so users can move from the overall fraud picture to specific risk and behavior segments.

The main business value of the project is that it connects fraud labels with actionable monitoring signals. The analysis identifies higher observed fraud rates in upper IP/device risk bands, Web transactions, Collect Request transactions and large location-change segments, while also highlighting a fraud-reason data-quality issue that should be resolved before automated use.

15. Skills Demonstrated

Area	Skills Demonstrated
Data Cleaning	Pandas profiling, datetime conversion, missing-value checks, duplicate checks, categorical validation, range checks and outlier investigation.
Python	Pandas aggregation, grouping, fraud-rate calculations, descriptive analysis and business-focused EDA.
SQL	COUNT, SUM, CASE, GROUP BY, fraud-rate calculations, segmentation and date/time analysis.
Power BI	Interactive dashboard design, KPI cards, slicers, charts, scatter plots and risk segmentation.
DAX	COUNTROWS, CALCULATE, SUM, AVERAGE, DIVIDE and conditional measures.
Business Analysis	Fraud KPI selection, risk-segment analysis, pattern identification, monitoring recommendations and data-quality interpretation.
Communication	Converting technical analysis into business problems, insights, recommendations and an interview-ready project story.

Project note: The supplied dataset is a project dataset. This report does not claim that it represents actual RBI/NPCI production transactions.